

Hadoop Execution Environment

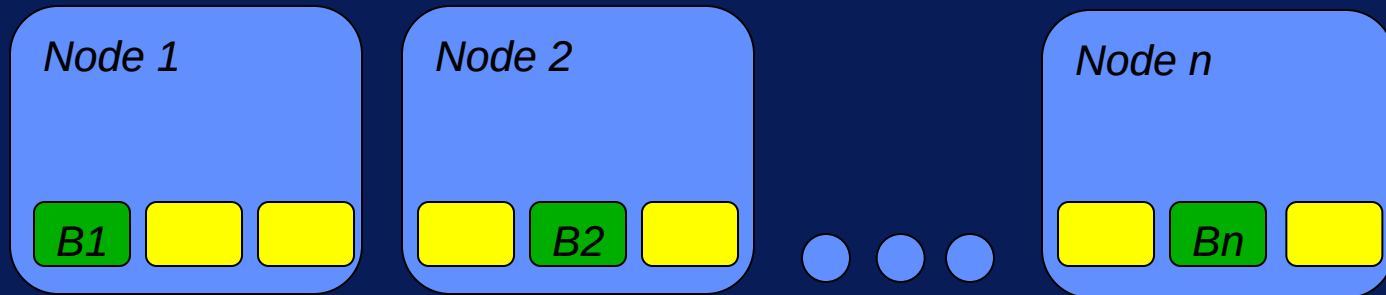


Hadoop Execution Environment

- *Learn about execution environments in Hadoop.*
- *Limitations of classic MapReduce framework.*
- *New frameworks like YARN, Tez, Spark to compliment classic MapReduce.*

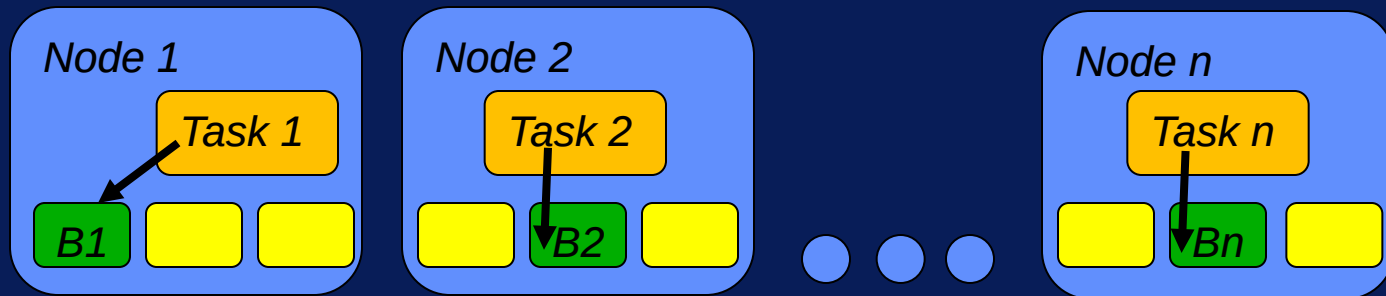
Recall Hadoop Architecture

- *Data distributed across nodes*



Recall Hadoop Architecture

- *Data distributed across nodes*
- *Keep compute task on the node with data.*

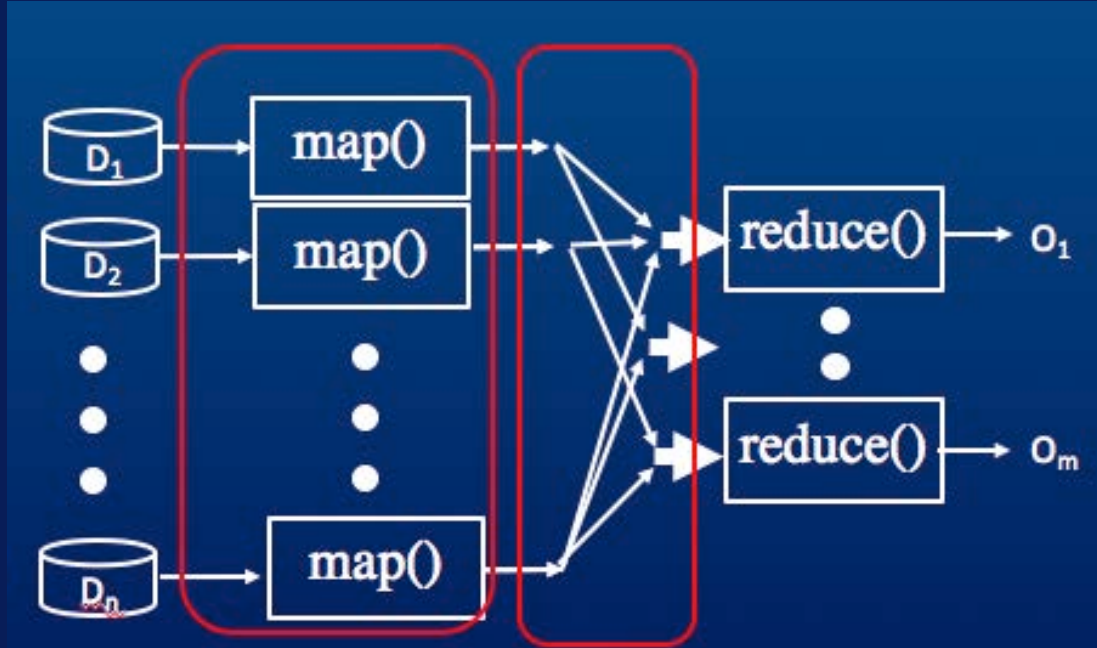


MapReduce Execution Framework

- Software framework
- *Schedules, monitors, and manages tasks*

MapReduce Execution Framework

- Works for Applications that fit MapReduce paradigm.*



NextGen Execution Frameworks

- *What if Application doesn't fit or is not efficient in MapReduce Paradigm?*

NextGen Execution Frameworks

- *What if Application doesn't fit or is not efficient in MapReduce Paradigm?*
 - *Interactive data exploration*
 - *Iterative data processing*

NextGen Execution Frameworks

- *Enter: Execution frameworks like YARN, Tez, Spark*
- *Support complex directed acyclic graph (DAG) of tasks.*
- *In memory caching of data*

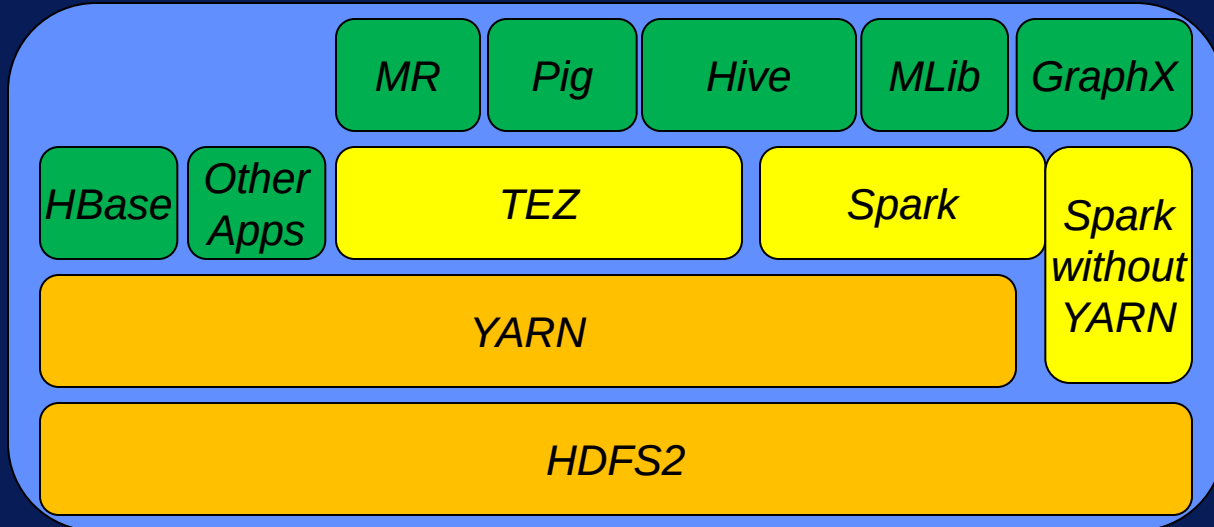
Lesson 2, Video #2

Hadoop Execution Environment

- *Layout of new frameworks (YARN, Tez, Spark) in Hadoop environment.*
- *Optimization strategies used in new frameworks.*
- *Examples illustrating use of Tez, Spark.*

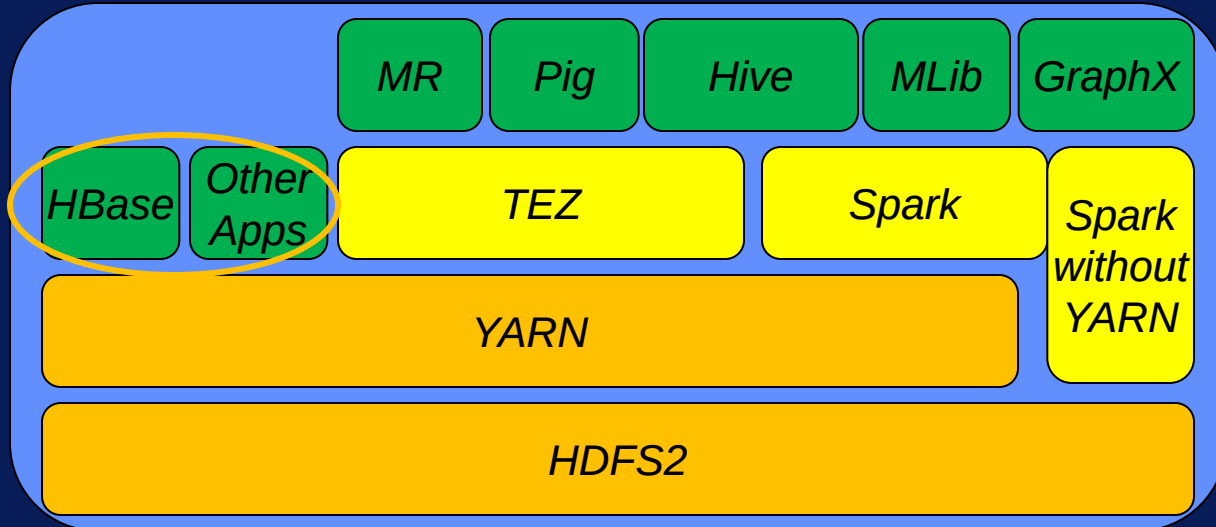
YARN, Tez, Spark

- Execution frameworks: YARN, Tez, and Spark



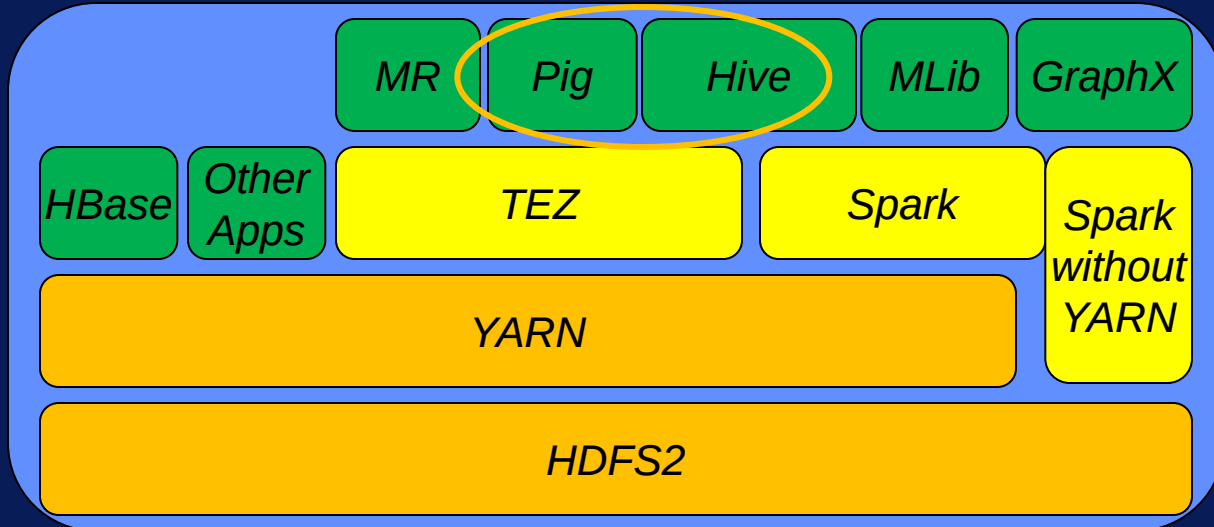
YARN, Tez, Spark

- Execution frameworks: YARN, Tez, and Spark



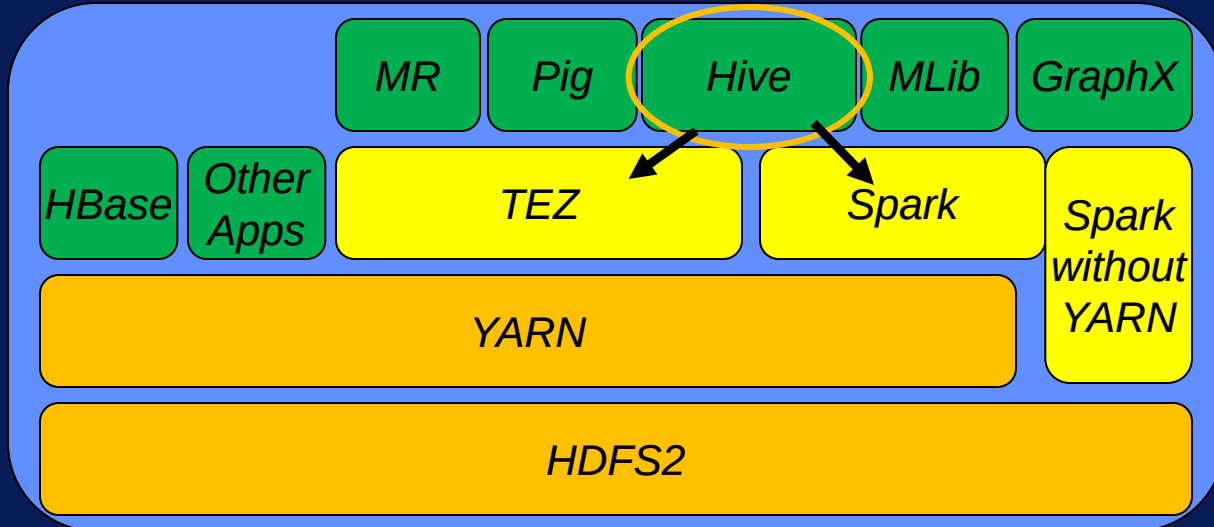
YARN, Tez, Spark

- Execution frameworks: YARN, Tez, and Spark



YARN, Tez, Spark

- Execution frameworks: YARN, Tez, and Spark



YARN

- *MapReduce*
- *Open source/commercial applications*
- *User developed applications*
- *Frameworks like Tez, Spark*

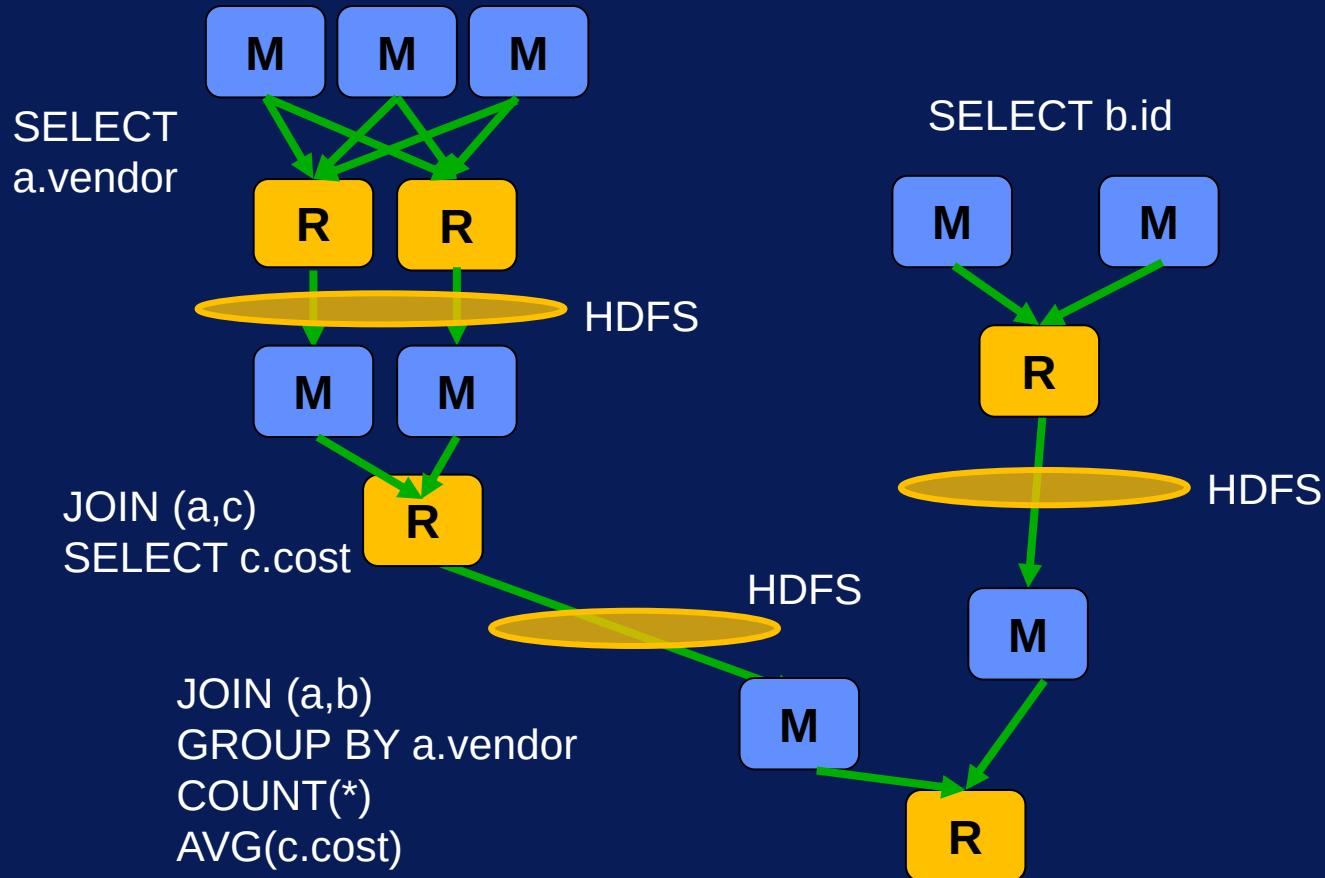
Tez

- *Dataflow graphs*
- *Custom data types*
- *Can run complex DAG of tasks*
- *Dynamic DAG changes*
- *Resource usage efficiency*

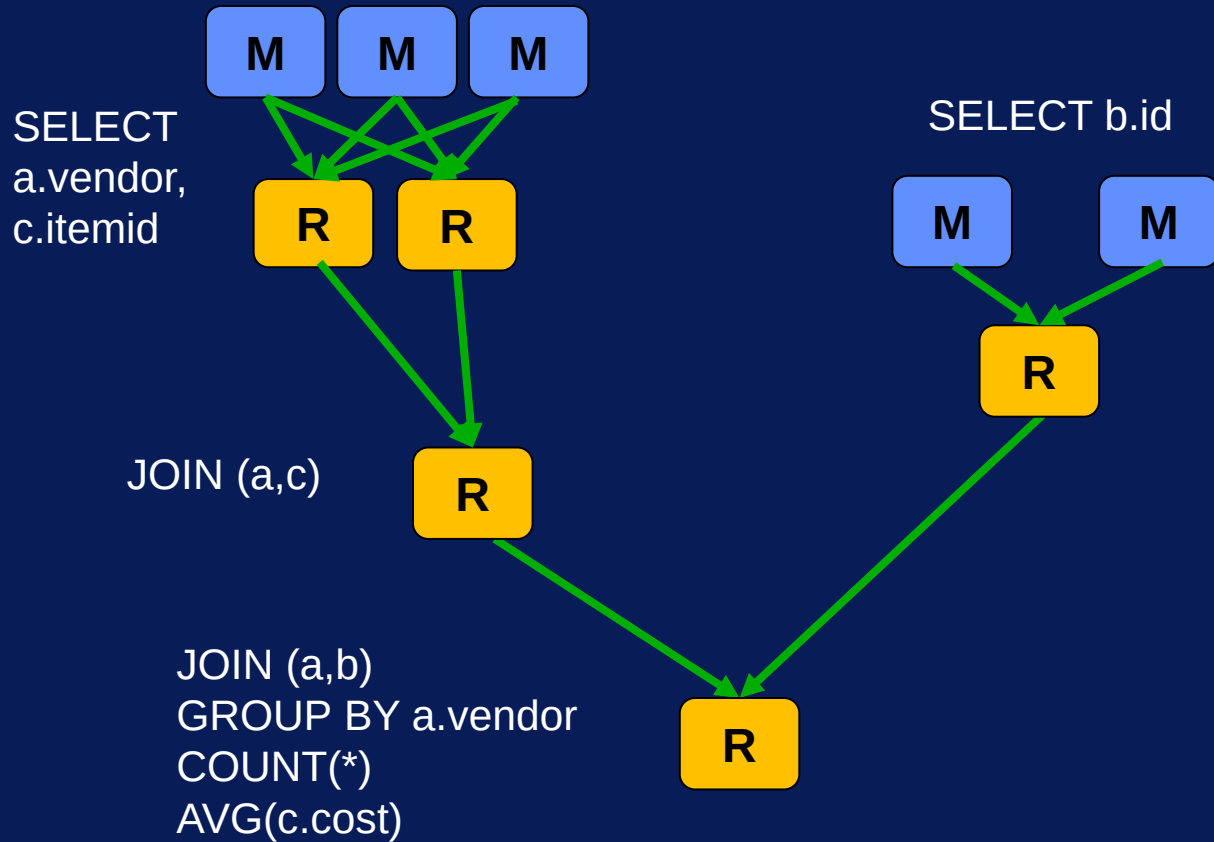
HIVE on Tez example

```
SELECT a.vendor, COUNT(*), AVG(c.cost) FROM a  
JOIN b ON (a.id = b.id)  
JOIN c ON (a.itemid = c.itemid)  
GROUP BY a.vendor
```

HIVE Example - MapReduce



HIVE Example - Tez



Spark

- *Advanced DAG execution engine*
- *Supports cyclic data flow*
- *In-memory computing*
- *Java, Scala, Python, R*
- *Existing optimized libraries*

Spark Example

- Logistic Regression example

```
points = spark.textFile(...).map(parsePoint).cache()
w = numpy.random.randn(size = D) # current separating plane
for i in range(ITERATIONS):
    gradient = points.map(
        lambda p: (1 / (1 + exp(-p.y*(w.dot(p.x)))) - 1) * p.y * p.x
    ).reduce(lambda a, b: a + b)
    w -= gradient
print "Final separating plane: %s" % w
```

Spark Example

- Logistic Regression example

```
points = spark.textFile(...).map(parsePoint).cache()
w = numpy.random.randn(size = D) # current separating plane
for i in range(ITERATIONS):
    gradient = points.map(
        lambda p: (1 / (1 + exp(-p.y*(w.dot(p.x)))) - 1) * p.y * p.x
    ).reduce(lambda a, b: a + b)
    w -= gradient
print "Final separating plane: %s" % w
```